

AI-Based Assembly Process Monitoring System

Beginner-Friendly Documentation

1. Project Overview

This project is a real-time computer vision system designed to monitor an industrial assembly process. It detects tools, components, and human actions using AI models and verifies whether each assembly step is performed in the correct order.

2. System Workflow

RTSP Camera → Video Frame Capture → YOLO Object Detection & Pose Estimation → Step Sequence Validation → Visual Overlay → Live Web Streaming via Flask

3. AI Models Used

Two models are used in this system:

- YOLO Object Detection Model: Detects parts, tools, drill machines, and hands.
- YOLO Pose Estimation Model: Used to understand human pose and hand interactions.

4. Detected Classes

Examples of detected classes include end cover, bolts, clamp, hammer, circlip, drain plug, loctite, cutoff valve, drill machine, and hand gloves. Some actions like tightening are derived using drill position logic.

5. What is an Assembly Cycle?

A cycle represents one complete and correct assembly operation. Each cycle must follow a predefined sequence of steps such as end cover fitment, bolt tightening, clamping, interlock operations, and final tightening.

6. Step Validation Logic

The system does not rely on a single frame. A step is considered completed only when the same object or action is detected consistently across multiple frames to avoid false positives.

7. Sequence Break Detection

If a step is detected out of order (for example, circlip detected before hammering), the system flags it as a sequence break and highlights it visually on the screen.

8. Drill Tightening Detection

Tightening actions are detected by tracking the drill machine position. If the drill stays within predefined tightening zones for a sufficient duration, the system confirms the tightening step.

9. Hand and Spring Detection

Hand detection is used to identify manual operations such as spring placement. Distance between both hands and their position inside defined regions determines whether the action is valid.

10. Visual Overlay

The live video feed displays:

- Current cycle status
- Completed steps (green)
- Pending steps (yellow)
- Sequence errors (red blinking)

11. Web Application

A Flask web server streams the processed video in real-time. Users can access the live feed through a browser and control recording or cycle reset.

12. Video Recording

The system supports recording. Each recording is saved with a unique filename to prevent overwriting previous files.

13. Multithreading

Multithreading allows video capture, AI inference, and web streaming to run smoothly without blocking each other. Thread locks ensure data safety.

14. Error Handling

The system automatically attempts to recover from RTSP disconnections, model inference errors, and empty frames.

15. Key Takeaway for Beginners

This project is not just object detection. It is an intelligent process verification system that understands industrial assembly workflows using AI.